

Electrical Architecture And Wiring Overview

1. High-Voltage Input (AC Side)

- **Source:** Standard 120V AC 3-prong wall plug.
- **Wiring:** * **Black (Hot):** Connects to the **L** (Line) terminal on the PSU.
 - **White (Neutral):** Connects to the **N** (Neutral) terminal on the PSU.
 - **Green (Ground):** Connects to the **G** (Ground) terminal on the PSU. **(Crucial for safety with a metal-housed PSU).**

2. Low-Voltage Distribution (12V DC Side)

- **PSU to Block:** Run a high-gauge wire from the **V+** and **V-** terminals of the PSU to the **Main Input** of the **Power Distribution Block**.
- **The Distribution Block:** This acts as the hub, ensuring each component gets a clean 12V feed without overloading the few screw terminals on the PSU.

3. Component Wiring:

- **Mixing motors:** Connect directly to the **PWM controllers**.
- **PWM Controllers:** Connect to the **Power Distribution Block** and **motors** to provide adjustable power and speed.
- **Cooling fans:** Connected directly to the **power distribution block**. - Always on to ensure the cabinet stays cool.
- **Solenoids:** Wired to the **power distribution block** and **momentary push buttons**
- **Push button:** Wired to the **power distribution block** and **solenoid**. When the button is pushed, power is supplied to the **solenoid** to open the valve.